



北京邮电大学



Queen Mary
University of London

QHM5701 A

Queen Mary School Hainan 2024/25

QHM5701 Linear Programming and Games

Paper A

Time allowed 3 hours

Answer ALL questions

Complete the information below yourself very carefully.

QM student number

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BUPT student number

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Class number

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NOT allowed: Any Electronic devices including electronic calculators and electronic dictionaries.

INSTRUCTIONS

1. **You must not take answer books, used or unused, from the examination room.**
2. Write only in black or blue pen **and in English.**
3. Do all rough work in the answer book – **do not tear out any pages**
4. If you use Supplementary Answer Book, tie them to the end of this book.
5. Write clearly and legibly.
6. **Read the instructions on the inside cover.**

Examiners

Dr Dimitris Chiotis, Dr Nivedita Viswanathan

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For examiners' use only

1	
2	
3	
4	
Total	

Instructions

Before the start of the examination

- 1) Place your BUPT and QM student cards on the corner of your desk so that your picture is visible.
- 2) Put all bags, coats and other belongings at the back/front of the room. All small items in your pockets, including wallets, mobile phones and other electronic devices must be **placed in your bag in advance. Possession of mobile phones, electronic devices and unauthorised materials is an offence.**
- 3) Please ensure your mobile phone is switched off and that no alarm will sound during the exam. **A mobile phone causing a disruption is also an assessment offence.**
- 4) Do not turn over your question paper or begin writing until told to do.

During the examination

- 1) You must not communicate with or copy from another student.
- 2) If you require any assistance or wish to leave the examination room for any reason, please raise your hand to attract the attention of the invigilator.
- 3) If you finish the examination early you may leave, but not in the first 30 minutes or the last 10 minutes.
- 4) For 2 hour examinations you may **not** leave temporarily.
- 5) For examinations longer than 2 hours you **may** leave temporarily but not in the first 2 hours or the last 30 minutes.

At the end of the examination

- 1) You must stop writing immediately — **if you continue writing after being told to stop, that is an assessment offence.**
- 2) Remain in your seat until you are told you may leave.

Question 1

- (a) Determine whether each of the following statements is True or False by ticking the respective box. You do not need to justify your choice. Each correct selection is awarded 1 mark.

Statement	True	False
1. A feasible solution to a linear program in standard inequality form satisfies all the constraints and sign restrictions.	☐	☐
2. An extreme point solution to a linear program is an internal point solution.	☐	☐
3. For a linear program in standard equation form, an extreme point solution is a basic feasible solution.	☐	☐
4. The feasible region is the set of all possible solutions that satisfy the constraints and the sign restrictions.	☐	☐
5. In all linear programs, every feasible solution is automatically an optimal solution.	☐	☐
6. Every 2-player zero sum game has a pure Nash Equilibrium.	☐	☐
7. For any primal linear program, it's dual linear program always has a unique basic feasible solution.	☐	☐
8. For feasible solutions x and y to be optimal for the primal and dual linear programs respectively, objective values at the points should be equal.	☐	☐
9. In a 2-player zero-sum game, the best pure strategy of a row player is the one that minimises his/her maximum payoff.	☐	☐
10. Suppose the given primal linear program has feasible solutions but an unbounded objective function. Then the dual has a unique feasible solution that is optimal.	☐	☐

[10 marks]

(b) Suppose that the following linear program is given:

$$\begin{aligned} & \text{maximize} && \mathbf{c}^\top \mathbf{x} \\ & \text{subject to} && A\mathbf{x} \leq \mathbf{b} \\ & && \mathbf{x} \geq \mathbf{0}, \end{aligned} \tag{1}$$

where $\mathbf{c}, \mathbf{x} \in \mathbb{R}^n$, $A \in \mathbb{R}^{m \times n}$ and $\mathbf{b} \in \mathbb{R}^m$.

[8 marks]

(i) Prove that if $\mathbf{x}_1$ and $\mathbf{x}_2$ are feasible solutions to problem 1, then

$$\mathbf{x} = \frac{1}{3}\mathbf{x}_1 + \frac{2}{3}\mathbf{x}_2$$

is also a feasible solution for (1).

(4 marks)

(ii) What does it mean to say that a feasible solution $\mathbf{y} \in \mathbb{R}^n$ is an optimal solution to the linear program (1)?

(2 marks)

(iii) When do we call such a linear program unbounded?

(2 marks)

(c) Consider the linear program

$$\begin{aligned} & \text{maximize} && \begin{pmatrix} 5 & 2 & -1 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} \\ & \text{subject to} && \begin{pmatrix} 1 & 1 & -1 & 4 \\ 1 & -2 & -1 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 8 \\ 2 \end{pmatrix} \\ & && \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix}^\top \geq \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}. \end{aligned}$$

For each of the vectors $\mathbf{x}_i^\top$ given below, determine whether or not $\mathbf{x}_i^\top$ is a **feasible solution** and also whether or not $\mathbf{x}_i^\top$ is a **basic feasible solution** to the given linear program. Justify your answers.

- $\mathbf{x}_1^\top = (6 \ 2 \ 0 \ 0)$
- $\mathbf{x}_2^\top = (3 \ 1 \ 0 \ 1)$
- $\mathbf{x}_3^\top = (-3 \ 0 \ 1 \ 2)$.

[4 marks]

Question 2

(a) Consider the following linear program:

$$\begin{array}{ll}\text{maximize} & 4x_1 + 3x_2 \\ \text{subject to} & x_1 + x_2 \leq 4 \\ & 5x_1 + 3x_2 \leq 15 \\ & x_1, x_2 \geq 0.\end{array}$$

[7 marks]

(i) Sketch the feasible region of the given linear program in two variables. You do not need to justify your plot, but make sure it is labeled.

(3 marks)

(ii) On the same plot, sketch the 2-level curve of the objective function, that is sketch the line $4x_1 + 3x_2 = 2$.

(1 marks)

(iii) Clearly indicate on your plot at which point of the feasible region the optimal solution of the linear program is attained. Please explain your answer.

(3 marks)

(b) Consider the following linear program:

$$\begin{aligned}
 &\text{minimize} && -25x_1 + 23x_2 - 50x_3 \\
 &\text{subject to} && x_1 + 4x_2 + 8x_3 \geq 15 \\
 &&& -x_1 + 3x_2 + 9x_3 \leq 25 \\
 &&& x_1 + x_3 = 7 \\
 &&& x_1, x_2, x_3 \geq 0.
 \end{aligned}$$

[5 marks]

(i) Convert the above linear program into its standard inequality form.

(2 marks)

(ii) Write the linear program in matrix notation.

(3 marks)

(c) Suppose we are given a linear programming problem and its standard equation form has variables $x_1, x_2, x_3 \geq 0$ and slack variables $s_1, s_2 \geq 0$. Also, suppose that after applying a few steps of the simplex algorithm we obtain the following tableau:

Basic	x_1	x_2	x_3	s_1	s_2	Solution
s_1	4	0	-4	1	-1	4
x_2	-3	1	2	0	1	3
$-z$	6	0	-3	0	-2	-6

[6 marks]

For this tableau:

(i) Identify the basic feasible solution and the basic variables at this step of the simplex.

(2 marks)

(ii) Identify the objective value at this step of the simplex.

(1 marks)

(iii) Suppose we want to proceed to the next tableau. Highlight the pivot column and pivot row of the given tableau above. You do not need to write the new tableau.

(3 marks)